

RAGA SIRI POTLURI

Data & Generative AI Engineer | Multi-Cloud Data Platforms (AWS, GCP, Azure) | Azure OpenAI, ETL, RAG, LLMs, Bedrock, Databricks, AWS Glue | Healthcare, Finance, Retail, Big Data Platforms
ragasiripotluri03@gmail.com | +1(945)-358-1178 | www.linkedin.com/in/raga-siri-potluri/

PROFESSIONAL SUMMARY

- **Data and Generative AI Engineer** with around **5** years of experience designing enterprise-scale **ETL** and **RAG** pipelines on **AWS**, delivering governed analytics and **LLM-Powered** solutions supporting 10+ business teams in **finance, retail, and healthcare domains**.
- Proven expertise building large-scale **data lakes** and **data warehouses** using **Amazon S3** and **Amazon Redshift**, processing over 1B+ records and 6TB daily to support analytics, reporting, and **AI-driven** decision systems.
- Strong background in credit risk and delinquency analytics, delivering **ETL** pipelines on **AWS Glue** and **Amazon Redshift** handling 25M monthly loan records across 12 products and 3K branches for regulatory and MIS reporting.
- Hands-on experience developing **RAG** architectures using **Amazon OpenSearch Service** and **Amazon Bedrock**, enabling semantic search and grounded summarization across 18M healthcare documents and serving 15K daily enterprise user queries.
- Expertise in **Spark** and **PySpark** for distributed batch processing, implementing 120M text chunk pipelines and 40+ complex metrics to support forecasting, risk analytics, and GenAI workloads at scale.
- Delivered supply chain and replenishment analytics on **AWS** using **AWS Glue**, **Amazon Redshift**, and **AWS Step Functions**, orchestrating 25 pipelines and supporting large-scale retail operations with SLA-driven inventory insights.
- Strong focus on data quality and governance, implementing 100+ validation rules and RBAC controls using **AWS IAM**, supporting 300+ users while ensuring audit-ready and compliant data access.
- Experienced in **CI/CD** pipeline orchestration and observability using **Amazon CloudWatch** and **AWS CodePipeline**, tracking 50+ operational metrics to ensure reliability, freshness, and performance across enterprise platforms.
- Demonstrated ability to optimize performance at scale, tuning **Amazon Redshift** tables exceeding 1B rows and optimizing **AWS Glue** jobs across multiple clusters to meet demanding analytics and GenAI workloads.
- Collaborative engineer with experience partnering with 20+ stakeholders including analysts, clinicians, and product teams, translating business rules into scalable **data engineering** and **Generative AI solutions** across 4+ regulated industries.

TECHNICAL SKILLS

Programming & Query Languages: Python, SQL, PySpark

Cloud Platforms: AWS, Azure, GCP

AWS Services: AWS S3, Glue, Lambda, AWS Redshift, AWS Bedrock, EMR, CloudWatch, OpenSearch, IAM, KMS, CodePipeline, CloudFormation

Azure Services: Azure Data Factory, Azure Databricks, Azure Data Lake Storage Gen2, Azure Synapse, Azure Monitor, Azure Purview, Power BI

GCP Services: BigQuery, Dataflow, Cloud Composer, Cloud Storage, Pub/Sub, Dataplex, Cloud Logging, Cloud Monitoring, Cloud Build

Data Engineering & Analytics: ETL, ELT, DBT, Data Lake, Data Warehouse, Lakehouse, Batch Processing, Streaming Pipelines, Data Modeling, Data Quality, Data Validation, Data Lineage, Metadata Management

Big Data & Processing: Spark, Delta Lake, BigQuery SQL, Distributed Processing, Partitioning, Clustering, Performance Tuning

Generative AI & ML: Gen AI, LLMs, RAG, Vector Search, Embeddings, Prompt Engineering, Semantic Search, Model Evaluation

Databases & Storage: BigQuery, Redshift, Snowflake, Delta Lake, OpenSearch, Cloud Storage, Azure Data Lake, Amazon S3

Orchestration & DevOps: Cloud Composer, Azure DevOps, AWS CodePipeline, CI/CD, Git, Infrastructure as Code , Monitoring, Alerting

Security & Governance: IAM, RBAC, KMS, VPC Service Controls, GxP Compliance, HIPAA, Regulatory Reporting, Audit Readiness

PROFESSIONAL EXPERIENCE

Data and Generative AI Engineer | Optum | Dallas, Texas

Jan 2025 - Present

Project: Built a secure, **AWS-native Generative AI assistant** enabling natural-language search, summarization, and analytics across healthcare claims, policy, and clinical documents using **Amazon S3**, **AWS Glue**, **Amazon OpenSearch Service**, and **Amazon Bedrock**.

- Designed end-to-end ETL and **RAG pipelines** using **AWS Glue** to ingest and normalize 18M healthcare documents from six source systems into **Amazon S3**, enabling scalable semantic retrieval and analytics-ready datasets.
- Implemented large-scale document preprocessing using **Spark** and **SQL** on **AWS Glue**, generating 120M text chunks with optimized token windows and overlap to improve retrieval accuracy across 25 claims and policy domains.
- Built vector indexing and enrichment workflows using **Amazon OpenSearch Service** storing 120M embeddings and enabling sub-second semantic queries across 3TB of unstructured healthcare content.
- Integrated **LLMs via Amazon Bedrock** into **RAG** workflows to generate grounded, context-aware responses for 15K daily enterprise queries, supporting claims adjudication, policy interpretation, and operational research across 10 business teams.
- Developed **SQL**-based metadata filtering logic combining vector similarity with structured attributes, improving precision for clinical, claims, and compliance analytics use cases.
- Orchestrated 12 production-grade **ETL** workflows using **AWS Glue** workflows and **AWS Step Functions**, ensuring daily ingestion of 2M new documents and embeddings with end-to-end SLAs under 90 minutes.
- Implemented **CI/CD** pipelines using **AWS CodePipeline** and **AWS CloudFormation** to automate deployment of Glue jobs, **OpenSearch** resources, and IAM policies across 3 environments, reducing manual releases and cutting promotion time by 40%.
- Enforced data quality and governance by implementing 80+ validation rules using **Spark** and **SQL**, covering completeness, schema drift, and duplicate detection across 2M daily ingested records in regulated healthcare workflows.
- Enabled secure and compliant access using **AWS IAM**, supporting 300+ internal users while ensuring HIPAA-aligned handling of PHI across analytics and GenAI workloads.
- Optimized **AWS Glue** jobs processing 6TB daily, reducing embedding generation runtimes by 32% while maintaining throughput for peak enterprise GenAI usage.
- Implemented monitoring and alerting using **Amazon CloudWatch**, tracking 40+ metrics including pipeline failures, retrieval depth, query latency, and **LLM** inference errors for proactive tuning.
- Delivered dashboards using **Amazon QuickSight** sourced from curated **Amazon S3** and **Amazon OpenSearch Service** datasets, providing visibility into document coverage, query trends, and pipeline SLAs.

- Partnered with clinicians, claims analysts, and compliance teams to translate business rules into retrieval strategies and prompt refinements, aligning **AWS-based GenAI** outputs with operational definitions across 4 healthcare programs.

Data Engineer | Sam's Club | Bentonville, Arkansas

May 2024 – Dec 2024

Project: Built an **AWS-native GenAI-enabled supply chain** and **replenishment analytics platform** processing 120M+ inventory, logistics, and supplier records per month, combining governed **ETL** pipelines with **LLM-powered insights** to support 2K retail locations, 15 distribution centers, and 8 supply chain teams with SLA-driven and AI-assisted decision making.

- Designed end-to-end **ETL** pipelines using **AWS Glue** to ingest and transform 120M supplier, shipment, and store inventory records per month from **Amazon S3** and streaming sources into **Amazon Redshift**, forming the analytical backbone for GenAI-driven supply insights.
- Integrated near-real-time ingestion using **Amazon Kinesis**, enabling incremental inventory updates and feeding streaming pipelines that support timely replenishment signals and exception detection.
- Built curated data warehouse layers in **Amazon Redshift** using **SQL-based transformations**, standardizing schemas across 15 distribution centers and 35 product categories to serve as trusted context for downstream LLM reasoning.
- Implemented scalable batch processing logic in **AWS Glue** to compute 40+ supply chain KPIs including inbound delays, fill rates, and stock-out metrics across 18M SKUs, ensuring consistent, high-quality inputs for GenAI analytics.
- Orchestrated 25 interdependent pipelines using **AWS Step Functions**, coordinating batch loads, streaming updates, and warehouse transformations with SLA tracking to deliver analytics-ready data within 75 minutes of arrival.
- Modeled analytics-ready fact and dimension tables in **Amazon Redshift** with 50+ attributes per SKU-location, enabling sub-second queries consumed by both dashboards and GenAI applications.
- Implemented **data governance** and **lineage** using **AWS Glue Data Catalog** and **AWS Lake Formation**, ensuring traceability, controlled access, and audit readiness for data used in LLM-powered insights.
- Developed robust data quality checks using **AWS Glue** and **SQL**, enforcing 120+ business rules across 120M rows per run to prevent low-quality data from propagating into GenAI outputs.
- Maintained 24 rolling monthly snapshots in **Amazon Redshift**, enabling historical trend analysis, seasonality detection, and back-testing of replenishment logic used by LLM-generated recommendations.
- Optimized **Amazon Redshift** tables exceeding 900M rows using distribution styles and sort keys, improving query latency by 32% and ensuring fast retrieval for real-time GenAI prompts.
- Integrated **Amazon Bedrock LLMs** with curated Redshift datasets to generate natural-language summaries explaining demand trends, stock risks, and replenishment exceptions for planners and operations leaders.
- Enabled conversational analytics by grounding **Bedrock-powered LLM** responses in real-time KPIs and historical context, reducing manual analysis effort for supply chain decision makers.
- Implemented **CI/CD** pipelines using **AWS CodePipeline** and **AWS CloudFormation** to automate deployment of Glue jobs, Step Functions workflows, Redshift schemas, and GenAI infrastructure.
- Enforced secure access using **AWS IAM** and private network controls, ensuring sensitive supplier and inventory data remained protected while supporting 40+ internal users across 14 roles.
- Implemented observability using **Amazon CloudWatch**, tracking 60+ metrics across 25 workflows and GenAI components to proactively detect data latency, pipeline failures, and insight degradation.
- Collaborated with 9 supply chain analysts and planners to translate operational rules into **ETL logic**, **KPI definitions**, and **LLM prompts**, aligning GenAI outputs with enterprise replenishment and lead-time definitions.

Project: Built an **AWS-native credit risk and delinquency analytics** platform processing high-volume lending data. Delivered governed, secure **ETL pipelines** on **AWS S3**, **AWS Glue**, and **AWS Redshift**, enabling portfolio risk tracking, early warning signals, and RBI-aligned MIS reporting at scale

- Designed end-to-end **ETL pipelines** on **AWS Glue** to ingest and transform 25M loan repayment and installment records per month from 5 core lending systems into **Amazon S3**, enabling centralized credit risk and delinquency analytics.
- Built **Amazon S3 data lake layers** with standardized schemas covering 12 loan products and 3K branches, supporting loan performance, overdue tracking, and 24 months of historical trend analysis.
- Implemented scalable transformations using **AWS Glue** and **PySpark** to compute 30+ delinquency indicators across 18M active accounts, ensuring consistent downstream risk metrics.
- Modeled analytics-ready fact and dimension tables in **Amazon Redshift** with 40+ attributes per loan, enabling **sub-second queries** for risk dashboards, portfolio reviews, and executive MIS across 10 business teams.
- Automated daily batch orchestration using **AWS Lambda** and **Amazon CloudWatch**, coordinating 15 dependent pipelines with SLA monitoring and delivering datasets within 90 minutes of source data arrival.
- Implemented comprehensive **data quality frameworks** enforcing 100+ business rules across 25M rows per run, supporting audit readiness, regulatory reporting, and analytics trust.
- Maintained 36 monthly rolling snapshots in **Amazon S3** to enable portfolio aging, cohort analysis, and risk back-testing across 8 lending segments.
- Optimized **Amazon Redshift** performance by tuning sort and distribution keys on tables exceeding 1.2B rows, improving analytics response times by 35% across 20 recurring reports.
- Integrated **Amazon CloudWatch** logs and metrics to monitor 50+ pipeline health indicators, enabling proactive alerts for delayed feeds, schema drift, and data volume anomalies across 15 batch workflows.
- Delivered **RBI-aligned MIS datasets** refreshed twice daily, supporting 6 regulatory and finance reporting processes with zero manual intervention.
- Enforced enterprise-grade security using **AWS IAM** and **AWS KMS**, ensuring least-privilege access and encryption of sensitive financial data across 12 roles and 30+ internal consumers.
- Implemented CI/CD and infrastructure governance using **AWS CodePipeline** and **AWS CloudFormation**, enabling repeatable deployment of Glue jobs, IAM policies, and monitoring resources.
- Collaborated with 7 risk analysts and product stakeholders to translate credit policy rules into governed AWS data transformations across 4 lending business units.

CERTIFICATION

- AWS Certified Data Engineer - Associate (DEA-C01)
- Microsoft Certified: Azure Data Engineer Associate (DP -203)
- Microsoft Certified: Azure AI Engineer Associate (AI -102)
- Google Professional Data Engineer (GCP-PDE)
- Databricks Certified Data Engineer Associate (PR000054)

EDUCATION

The University of Texas at Dallas - Richardson, USA

Master of Science in Computer Science

GPA: 3.6/4